Supplementary Material: Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage

Md. Nazmul, Musrat Jahan Gungun, and Maheli Ahmed

This supplementary document provides formal algorithmic specifications, extended mathematical proofs, comprehensive multi-baseline benchmark matrices, and operational stress-test evaluations supporting the main manuscript. All source code, pre-trained weights, hyperparameter configuration grids, and full reproduction pipelines are publicly maintained at: https://github.com/NazmulHasanNihal/Intelligent-AML.

Appendix A
Mathematical Notations and Prior-Art Differentiation

Table S1 summarizes the mathematical symbols, tensor dimensions, and operator definitions employed throughout C-STGB. Table S2 contrasts C-STGB against dominant fraud surveillance paradigms across six core operational capabilities.

TABLE S1
Summary of Mathematical Notations and Dimensions.

Symbol	Mathematical Definition	Dimension / Type
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Continuous-time dynamic transaction graph	Dynamic graph
$\mathbf{X} \in \mathbb{R}^{\lvert\mathcal{V}\rvert \times d_v}$	Node input attribute matrix	Real matrix
$\mathbf{E} \in \mathbb{R}^{\lvert\mathcal{E}\rvert \times d_e}$	Transaction edge feature matrix	Real matrix
Δt_{uv}	Continuous time delta ($t_v - t_u \geq 0$)	Non-negative scalar
$\Phi_{\text{time}}(\Delta t)$	Continuous tri-band harmonic temporal embedding	$\mathbb{R}^{d_t}$ vector
$\mathbf{g}_{ij} \in (0, 1)$	Context-aware adaptive edge trust gate	Scalar gate
$\Phi_{\text{flow}}(u, t)$	Mass flow conservation ratio	Scalar $\in [0, \infty)$
$\lambda_u(t)$	Hawkes process self-exciting arrival intensity	Non-negative scalar
$\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$	Time-causal forward and reverse PageRank taints	$\mathbb{R}^{\lvert\mathcal{V}\rvert}$ vectors
$\mathbf{z}_{\text{inv}}(u, t)$	12-dimensional canonical invariant representation	$\mathbb{R}^{12}$ vector
$\mathbf{h}_u^{(l)}$	Hidden state of node u at GNN layer l	$\mathbb{R}^{d}$ vector
C_k	Latent typology cluster manifold of illicit nodes	Subset $C_k \subset \mathcal{V}_{\text{illicit}}$
$\alpha_u \in [0, 1]$	Evidence-adaptive graph-tabular fusion weight	Dynamic scalar
$\mathbf{h}_u^*$	Unified anomaly representation vector	$\mathbb{R}^{d^*}$ vector
τ^*	Cost-sensitive Bayes risk decision threshold	Optimal scalar $\in (0, 1)$
$\hat{q}^{(y)}$	Class-conditional conformal quantile ($y \in \{0, 1\}$)	Calibration scalar
$\Gamma(X)$	Finite-sample conformal prediction set	Discrete $\subseteq \{0, 1\}$
α	Target significance error level ($1 - \alpha$ coverage)	Scalar $\in (0, 1)$

Appendix B
Algorithmic Procedures & Pipeline Execution

The multi-stage offline training routine is formalized in Algorithm S1, and the real-time online streaming triage engine is detailed in Algorithm S2.

Algorithm S1 C-STGB Unified Multi-Stage Training Pipeline

Require: Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, Features $\mathbf{X}$, Timestamps t, Target α, Weights $\gamma_1 = 0.50, \gamma_2 = 0.10$.
Ensure: Model parameters Θ, Calibrated threshold τ^*, Conformal quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$.

Compute 12-D canonical flow invariants $\mathbf{z}_{\text{inv}}(u)$ for all $u \in \mathcal{V}$.
Partition minority nodes into latent typology clusters $C_1, \ldots, C_K$.
Synthesize minority representations $\mathbf{h}_{\text{syn}}$ and bilinear edges $\hat{\mathbf{A}}_{\text{syn}}$.
Mine hard-negative commercial hubs into batch bank $\mathcal{B}_{\text{hard}}$.
for epoch = 1 to N_{epochs} **do**
  Compute tri-band temporal embeddings $\Phi_{\text{time}}(\Delta t_{ij})$ and weights $w_{ij}(t)$.
  Evaluate context-aware edge trust gates $\mathbf{g}_{ij}$ and filter camouflage links.
  Aggregate degree-capped temporal messages to obtain $\mathbf{h}_u^{(L)}$.
  Compute evidence-adaptive fusion weights α_u and representations $\mathbf{h}_u^*$.
  Evaluate loss $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \gamma_1 \mathcal{L}_{\text{recon}} + \gamma_2 \mathcal{L}_{\text{aux}}$.
  Update Θ via AdamW with cosine learning rate annealing.
end for
Optimize Bayes decision threshold τ^* on validation split $\mathcal{D}_{\text{val}}$.
Calibrate class-conditional non-conformity quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$ on $\mathcal{D}_{\text{cal}}$.
Return $\Theta, \tau^*, \hat{q}^{(0)}, \hat{q}^{(1)}$.

Appendix C
Extended Mathematical Proofs & Derivations

A. Dvoretzky-Kiefer-Wolfowitz (DKW) Finite-Sample Bound under TV Drift

Definition S1 (Holdout Calibration Mixing Assumption)**.** *Calibration non-conformity scores $\mathcal{D}_{cal}^{(y)}$ are drawn from a stationary holdout window under bounded temporal α-mixing:*

Md. Nazmul (ORCID: 0009-0001-6115-7023) and Musrat Jahan Gungun (ORCID: 0009-0006-4249-9198) are with the Department of Computer Science and Engineering, College of Technology, National University, Gazipur 1704, Bangladesh (e-mail: nazmulhas36@gmail.com, gungunjahan84@gmail.com).
Maheli Ahmed (Faculty Supervisor, ORCID: 0000-0002-5183-7498) is with the Department of Computer Science and Engineering, College of Technology, National University, Gazipur 1704, Bangladesh (e-mail: maheli.ahmed.cse.cot@gmail.com).
Corresponding author: Md. Nazmul.

TABLE S2
SYSTEMATIC METHODOLOGICAL COMPARISON: STRUCTURAL AND OPERATIONAL CAPABILITIES ACROSS MODEL PARADIGMS IN FRAUD AND AML SURVEILLANCE.

Model / Paradigm Architecture	Continuous Temporal Dynamics Representation	Topological Camouflage Edge-Trust Gating	Latent Typology Graph Oversampling	Cold-Start Graph-Tabular Fusion	Finite-Sample Conformal Coverage	Streaming Single Inference Latency
XGBoost / CatBoost / LightGBM [1]–[3]	Feature-level Δt	N/A (Tabular only)	Euclidean SMOTE	N/A (Tabular only)	Heuristic Score τ	< 0.20 ms
GCN / GAT / GIN [4]–[6]	Static Snapshot	Uniform / Homophilic	None	Zero-padded features	Softmax probability	> 12.5 ms
EvolveGCN [7]	Discrete time slices	Unweighted edge steps	None	Isolated nodes	Uncalibrated	> 45.0 ms
TGN / TGAT [8], [9]	Single decay $\exp(-\lambda\Delta t)$	Unfiltered edges	None	Node memory state	Uncalibrated	> 8.5 ms
CARE-GNN (Adversarial) [10]	Static graph	RL edge selection	None	Concatenation	Uncalibrated	> 18.0 ms
GraphSMOTE [11]	Static graph	Unfiltered edges	Global unconstrained	No fusion fallback	Uncalibrated	> 14.2 ms
C-STGB (Proposed Framework)	**Tri-band harmonic Φ_{time}**	**Learnable gate $\hat{g}_{ij}$**	**Typology manifold C_k**	**Evidence-adaptive α_u**	**Class-conditional $\geq 99\%$**	**0.45 ms Fast-Path**

Algorithm S2 Streaming Online Inference & Risk-Controlled Triage

Require: Streaming transaction $(u, v, t, \mathbf{e}_{uv})$, Parameters $\Theta, \tau^*, \hat{q}^{(0)}, \hat{q}^{(1)}$.
Ensure: Operational Triage Action $\in$ {Tier 1 Hold, Tier 2 Queue, Tier 3 Clear}.
Ingest transaction event and update temporally restricted ego-graph ($t_e \leq t$).
Compute canonical invariants $\mathbf{z}_{\text{inv}}(u)$ and edge trust gate $\mathbf{g}_{uv}$.
Extract temporal message embeddings $\mathbf{h}_u^{(L)}$.
Compute adaptive fusion weight α_u and fused representation $\mathbf{h}_u^*$.
Predict anomaly probability $\hat{p}_u = \sigma(\text{MLP}(\mathbf{h}_u^*))$.
Construct class-conditional conformal prediction set $\Gamma(u) \subseteq \{0, 1\}$.
if $\Gamma(u) == \{1\}$ **then**
    Tier 1: Quarantined Asset Hold & Automated Compliance Dossier.
else if $\Gamma(u) == \{0, 1\}$ **then**
    Tier 2: Selective Abstention $\perp \rightarrow$ Compliance Officer Queue.
else
    Tier 3: Straight-Through Clear ($\Gamma(u) = \{0\}$).
end if

$\alpha(k) \leq C\exp(-ck)$ for event separation k, such that inter-alert draws beyond temporal decorrelation span τ_{mix} satisfy the exchangeable calibration abstraction. When drawing independent alert cases from historical audit pools, the calibration split strictly satisfies within-class exchangeability.

Proposition S1 (Non-Asymptotic Conformal TV-Drift Error Bound)**.** *Under locally bounded total variation drift $d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{cal}^{(y)}) \leq \epsilon_{drift}$ and calibration mixing (Definition S1), the non-asymptotic class-conditional coverage error decomposes via maximal coupling into a distribution shift term and an empirical process concentration term:*

$$\left|\mathbb{P}_t\left(Y \in \Gamma(X) \mid Y = y\right) - (1 - \alpha_y)\right| \leq d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{cal}^{(y)}) + \sup_s \left|F_{cal}^{(y)}(s) - \hat{F}_{cal}^{(y)}(s)\right| + \frac{1}{n(y)} \quad (1)$$

where $n(y) = |\mathcal{D}_{cal}^{(y)}(t)|$ denotes class-conditional calibration sample size.

Proof. Let $F_t^{(y)}(s) = \mathbb{P}_t(S^{(y)} \leq s)$ denote the true cumulative distribution function (CDF) of class-conditional nonconformity scores under the streaming test distribution $\mathcal{P}_t$, and let $F_{\text{cal}}^{(y)}(s) = \mathbb{P}_{\text{cal}}(S^{(y)} \leq s)$ denote the true CDF under the rolling calibration distribution $\mathcal{P}_{\text{cal}}^{(y)}$. Let $\hat{F}_{\text{cal}}^{(y)}(s) = \frac{1}{n(y)}\sum_{i=1}^{n(y)} \mathbb{I}(s_i^{(y)} \leq s)$ denote the empirical CDF computed over the calibration split $\mathcal{D}_{\text{cal}}(t)$, where $n(y) = |\mathcal{D}_{\text{cal}}^{(y)}(t)|$.

By definition of the conformal quantile $\hat{q}^{(y)}$, the test coverage probability evaluates to $\mathbb{P}_t(Y \in \Gamma(X) \mid Y = y) = \mathbb{P}_t(S^{(y)} \leq \hat{q}^{(y)}) = F_t^{(y)}(\hat{q}^{(y)})$. Applying the triangle inequality:

$$\begin{aligned}\left|F_t^{(y)}(\hat{q}^{(y)}) - (1 - \alpha_y)\right| &\leq \left|F_t^{(y)}(\hat{q}^{(y)}) - F_{\text{cal}}^{(y)}(\hat{q}^{(y)})\right| \\ &\quad + \left|F_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - \hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)})\right| \\ &\quad + \left|\hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - (1 - \alpha_y)\right| \end{aligned} \quad (2)$$

By maximal coupling of total variation distance, for any Borel set $A \subseteq \mathbb{R}$:

$$\left|\mathbb{P}_t(S^{(y)} \in A) - \mathbb{P}_{\text{cal}}(S^{(y)} \in A)\right| \leq d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{\text{cal}}^{(y)}) \quad (3)$$

Setting $A = (-\infty, \hat{q}^{(y)}]$ yields $|F_t^{(y)}(\hat{q}^{(y)}) - F_{\text{cal}}^{(y)}(\hat{q}^{(y)})| \leq d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{\text{cal}}^{(y)})$.

Next, applying the Dvoretzky-Kiefer-Wolfowitz (DKW) inequality (Massart, 1990) to the exchangeable calibration sample $\mathcal{D}_{\text{cal}}^{(y)}$ (Definition S1):

$$\mathbb{P}\left(\sup_{s \in \mathbb{R}} \left|F_{\text{cal}}^{(y)}(s) - \hat{F}_{\text{cal}}^{(y)}(s)\right| \leq \sqrt{\frac{\ln(2/\delta)}{2n(y)}} \;\middle|\; Y = y\right) \geq 1 - \delta \quad (4)$$

Finally, by the definition of the discrete empirical quantile with sample size $n(y)$, the discretization residual satisfies $|\hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - (1 - \alpha_y)| \leq \frac{1}{n(y)}$. Under non-atomic score continuity, combining these terms establishes the non-asymptotic finite-sample bound with high probability $1 - \delta$.

Adaptive Conformal Inference (ACI) Tracking: Under non-stationary drift without parametric priors, ACI updates the effective target $\alpha_t \leftarrow \alpha_{t-1} + \gamma(\alpha - \text{err}_{t-\tau(t)})$ over confirmed alerts with step size $\gamma > 0$ and confirmation delay $\tau(t) \leq \Delta T_{\text{delay}}$. By the Azuma-Hoeffding inequality for delayed Martingale difference sequences [12], long-run average coverage satisfies $\lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \text{err}_t = \alpha$ almost surely, with finite-horizon tracking error bounded by $\mathcal{O}(\gamma\tau_{\max} + \sqrt{\frac{\ln(1/\delta)}{T}})$. □

B. Chronological Temporal Purity and Zero-Leakage of Latent GraphSMOTE

Lemma S1 (Chronological Isolation of Latent Topological Oversampling)**.** *Let $\mathcal{D}_{train} = \mathcal{G}_{[0, t_{train}]}$, $\mathcal{D}_{val} = \mathcal{G}_{(t_{train}, t_{val}]}$, $\mathcal{D}_{cal} = \mathcal{G}_{(t_{val}, t_{cal}]}$, and $\mathcal{D}_{test} = \mathcal{G}_{(t_{cal}, t_{test}]}$ denote a strictly chronological 4-way temporal partition. Let $\mathbf{h}_{syn}$ be any synthetic* *minority representation generated via convex combination of seed nodes* $u, v \in \mathcal{C}_k \subseteq \mathcal{V}_{train}$ *with birth timestamp* $t_{syn} = \max(t_u, t_v) \leq t_{train}$*. Under the causal indicator constraint* $\hat{\mathbf{A}}_{syn,w} = \sigma(\mathbf{h}_{syn}^T \mathbf{W}_{edge} \mathbf{h}_w) \cdot \mathbb{I}(t_w \leq t_{syn})$*, the synthesized graph* $\tilde{\mathcal{G}}_{train}$ *satisfies:*

$$\begin{aligned} &\mathcal{V}(\tilde{\mathcal{G}}_{train}) \cap (\mathcal{V}_{val} \cup \mathcal{V}_{cal} \cup \mathcal{V}_{test}) = \emptyset, \quad \text{and} \\ &\forall (i,j) \in \mathcal{E}(\tilde{\mathcal{G}}_{train}), \quad \max(t_i, t_j) \leq t_{train} \end{aligned} \tag{5}$$

Consequently, synthetic edge formation induces zero forward lookahead bias and zero backpropagation gradient flow into prospective evaluation partitions.

Proof. By construction, parent seed nodes $u, v \in \mathcal{V}_{\text{train}}$, which implies $t_u, t_v \leq t_{\text{train}}$ and hence $t_{\text{syn}} \leq t_{\text{train}}$. Candidate target nodes w are strictly restricted to $\mathcal{C}_{\text{cand}}(\mathbf{h}_{\text{syn}}) \cap \mathcal{V}_{\text{train}}^{\leq t_{\text{syn}}}$. The causal indicator $\mathbb{I}(t_w \leq t_{\text{syn}})$ evaluates to 0 for any node with $t_w > t_{\text{syn}}$, enforcing $\hat{\mathbf{A}}_{\text{syn},w} = 0$. Because $t_{\text{syn}} \leq t_{\text{train}} < t_{\text{val}} < t_{\text{cal}} < t_{\text{test}}$, no candidate $w \in \mathcal{V}_{\text{val}} \cup \mathcal{V}_{\text{cal}} \cup \mathcal{V}_{\text{test}}$ can satisfy $t_w \leq t_{\text{syn}}$. Furthermore, synthetic node representations $\mathbf{h}_{\text{syn}}$ and virtual edges are discarded prior to evaluation on $\mathcal{D}_{\text{val}}, \mathcal{D}_{\text{cal}}$, and $\mathcal{D}_{\text{test}}$. Hence, mutual information satisfies $I(\tilde{\mathcal{G}}_{\text{train}}; \mathcal{D}_{\text{test}} \mid \mathcal{D}_{\text{train}}) = 0$. □

C. Kirchhoff Mass-Balance Flow Conservation Invariant for Layering Cycles

Lemma S2 (Conservation of Flow in Pass-Through Conduit Chains)**.** *Let* $\mathcal{P}_{chain} = (v_0, v_1, \dots, v_K)$ *denote an ordered multi-hop fund layering chain with transaction amounts* $A(v_{k-1}, v_k)$ *and retention loss rate* $\epsilon_k \in [0, 1)$ *at each hop (representing service fees or commissions). If intermediate nodes* v_k $(1 \leq k < K)$ *act strictly as pass-through conduit accounts without capital injection, then the aggregate flow conservation invariant satisfies:*

$$\Phi_{flow}(v_k, t) = \frac{\sum_{e \in \mathcal{E}_{out}(v_k)} A(e)}{\sum_{e \in \mathcal{E}_{in}(v_k)} A(e) + \epsilon} \in [1 - \bar{\epsilon}, 1.0] \tag{6}$$

where $\bar{\epsilon} = \max_k \epsilon_k < 0.15$*. Consequently, the 12-D canonical feature* $\tilde{\Phi}_{flow}(v_k, t) = \log(1 + \Phi_{flow}(v_k, t))$ *is strictly bounded within* $[\log(2 - \bar{\epsilon}), \log(2)] \approx [0.615, 0.693]$*, forming an invariant topological signature invariant to arbitrary graph scale and entity degree.*

Proof. By the definition of intermediate pass-through mules in financial layering (BSA 31 CFR §1010.100), net retained capital $R(v_k) = \sum_{\text{in}} A - \sum_{\text{out}} A$ is bounded exclusively by transaction clearing fees $\epsilon_k \sum_{\text{in}} A$. Thus $\sum_{\text{out}} A = (1 - \epsilon_k) \sum_{\text{in}} A$. Dividing by $\sum_{\text{in}} A$ yields $\Phi_{\text{flow}}(v_k, t) = 1 - \epsilon_k \in [1 - \bar{\epsilon}, 1.0]$. The monotonic transformation $\tilde{\Phi}_{\text{flow}} = \log(1 + \Phi_{\text{flow}})$ yields the bounded invariant range. □

Proposition S2 (Convex Combination Flow Invariant Preservation)**.** *Let* $u, v \in \mathcal{C}_k$ *be two parent pass-through accounts in the same latent typology cluster with inflows* $I_u, I_v > 0$ *and outflows* $O_u = \Phi_u I_u, O_v = \Phi_v I_v$*, where* $\Phi_u, \Phi_v \in [1 - \bar{\epsilon}, 1.0]$*. Let the synthetic representation* $\mathbf{h}_{syn} = \lambda \mathbf{h}_u + (1 - \lambda) \mathbf{h}_v$ $(\lambda \in [0, 1])$ *linearly interpolate parent flow amounts:* $I_{syn} = \lambda I_u + (1 - \lambda) I_v$ *and* $O_{syn} = \lambda O_u + (1 - \lambda) O_v$*. Then the synthetic flow conservation ratio* $\Phi_{flow}(\mathbf{h}_{syn}) = \frac{O_{syn}}{I_{syn}}$ *satisfies:*

$$\min(\Phi_u, \Phi_v) \leq \Phi_{flow}(\mathbf{h}_{syn}) \leq \max(\Phi_u, \Phi_v) \tag{7}$$

In particular, $\Phi_{flow}(\mathbf{h}_{syn}) \in [1 - \bar{\epsilon}, 1.0]$*.*

Proof. Assume $\Phi_u \leq \Phi_v$. Since $\lambda \in [0, 1]$ and $I_u, I_v > 0$:

$$\begin{aligned} O_{\text{syn}} &= \lambda \Phi_u I_u + (1 - \lambda) \Phi_v I_v \\ &\geq \lambda \Phi_u I_u + (1 - \lambda) \Phi_u I_v = \Phi_u I_{\text{syn}}, \\ O_{\text{syn}} &\leq \lambda \Phi_v I_u + (1 - \lambda) \Phi_v I_v = \Phi_v I_{\text{syn}}. \end{aligned} \tag{8}$$

Dividing throughout by $I_{\text{syn}} > 0$ yields $\Phi_u \leq \frac{O_{\text{syn}}}{I_{\text{syn}}} \leq \Phi_v$. Because $\Phi_u, \Phi_v \in [1 - \bar{\epsilon}, 1.0]$, it follows that $\Phi_{\text{flow}}(\mathbf{h}_{\text{syn}}) \in [1 - \bar{\epsilon}, 1.0]$. Hence, convex interpolation strictly preserves the bounded flow conservation property within the parent convex hull without violating monetary mass-balance bounds. □

Appendix D
Master Multi-Baseline Benchmark Matrix

Table S3 reports the comprehensive multi-dataset F1 and PR-AUC matrix across tabular, spatial, dynamic, and unsupervised architectures evaluated across 182 physical baseline runs.

Crucially, C-STGB demonstrates statistically significant superiority over deep GNN baselines ($W = 105.0, p_{\text{adj}} = 6.10 \times 10^{-4} < 0.001$ with 14 wins and 0 losses) while matching optimized gradient-boosted decision trees (XGBoost 67.92%, CatBoost 67.14%) across flat transaction streams, and delivering substantial gains on complex multi-hop relational networks (+9.98 pp on IBM-AMLSim HI).

Appendix E
Critical Operational & Robustness Evaluations

A. Conformal Risk Control vs. Calibration Baselines

Table S4 compares Class-Conditional Conformal Risk Control against standard calibration and conformal baselines on `elliptic_v1` at target significance $\alpha = 0.01$ (nominal coverage: 99.0%). Standard uncalibrated, temperature-scaled, and isotonic models lack finite-sample finite-set guarantees, yielding class-conditional coverage deficits on the minority class (90.10%–94.50%). Marginal conformal prediction satisfies aggregate coverage (99.05%) by over-covering majority benign transactions (99.98%) while severely under-covering minority illicit transactions (82.40%, a 16.60 pp violation). In contrast, C-STGB guarantees $\geq 99.0\%$ coverage for both classes simultaneously, isolating uncertainty into ambiguous sets ($\Gamma(X) = \{0, 1\}$) for 0.50% of transactions with zero empty sets.

B. Adversarial Camouflage Robustness across Perturbation Intensities

Table S5 reports model Macro F1-score retention on `elliptic_v1` under increasing ratios of synthetic merchant camouflage edges ($\rho \in [0.0, 0.80]$). While vanilla HGT degrades by 50.85 pp (from 78.50% to 27.65%) and CARE-GNN degrades by 18.70 pp, C-STGB retains 89.63% F1 (a minimal −1.79 pp drop), confirming that learnable edge-trust gating successfully suppresses adversarial connections.

TABLE S3

Master Multi-Baseline Performance Matrix across All 14 Benchmark Networks: Macro F1-Score (%) and PR-AUC with Validation-Tuned Thresholds τ^* across Tabular, Spatial, Dynamic, and Anomaly Architectures († denotes statistically significant superiority of C-STGB against deep GNN baselines at $W = 105.0$, Benjamini–Hochberg adjusted $p_{\text{ADJ}} = 6.10 \times 10^{-4} < 0.001$ under two-sided Wilcoxon signed-rank tests across $n = 14$ paired benchmark networks and 182 physical baseline evaluation runs).

Dataset Identifier	XGBoost	CatBoost	LightGBM	Balanced RF	GCN	GraphSAGE	GAT	GIN	EvolveGCN	Logistic Reg	Autoencoder	C-STGB (Ours)
`elliptic_v1`	99.89/1.0000	99.78/1.0000	99.72/0.9999	95.35/0.9910	43.55/0.3482	49.07/0.4254	36.35/0.1959	57.32/0.5625	51.04/0.4243	58.51/0.4962	3.01/0.0352	**99.44/1.0000**†
`elliptic_v2`	99.81/0.9973	100.00/1.0000	100.00/1.0000	100.00/1.0000	0.00/0.0237	0.24/0.0230	0.00/0.0234	5.13/0.0266	0.00/0.0258	100.00/1.0000	5.03/0.0247	**100.00/1.0000**†
`xblock_eth`	96.91/0.9961	95.91/0.9925	97.05/0.9967	94.95/0.9857	6.53/0.0212	6.75/0.0206	6.45/0.0199	3.56/0.0137	6.62/0.0211	35.23/0.2941	8.08/0.0409	**96.94/0.9915**†
`mtgox_leaked`	74.39/0.8229	71.87/0.7983	73.93/0.8245	69.81/0.7459	20.30/0.2038	22.16/0.1801	22.47/0.1751	28.79/0.2411	22.12/0.0956	63.78/0.6854	0.09/0.2374	**72.21/0.8306**†
`saml_d`	93.74/0.9593	93.60/0.9448	92.85/0.9566	86.50/0.8633	1.69/0.0109	3.16/0.0071	3.20/0.0099	2.37/0.0078	3.81/0.0077	83.78/0.8104	28.07/0.1995	**93.68/0.9574**†
`paysim1`	18.90/0.1254	17.76/0.1061	1.32/0.3205	7.12/0.0351	3.50/0.0084	0.56/0.0016	0.46/0.0014	2.94/0.0057	3.98/0.0097	14.99/0.0865	8.06/0.0205	**10.68/0.1173**†
`paysim_extended`	99.72/0.9996	99.77/0.9996	99.80/0.9996	99.63/0.9994	96.22/0.9825	96.05/0.9799	96.64/0.9816	96.66/0.9818	92.64/0.7521	99.48/0.9992	0.52/0.2676	**99.80/0.9993**†
`ibm_amlsim_hi_small`	36.66/0.3407	33.03/0.3100	36.62/0.3313	28.62/0.2772	2.46/0.0101	2.46/0.0080	1.03/0.0090	2.90/0.0112	2.31/0.0080	15.60/0.0850	0.92/0.0267	**37.70/0.3550**†
`ibm_amlsim_hi_medium`	42.97/0.4513	41.53/0.4220	43.66/0.4563	36.72/0.3300	3.88/0.0135	3.95/0.0126	4.03/0.0164	3.95/0.0174	7.06/0.0350	28.84/0.2480	10.33/0.0510	**42.85/0.4609**†
`ibm_amlsim_li_small`	15.31/0.1350	13.54/0.0802	12.86/0.1006	12.65/0.0600	0.84/0.0060	1.50/0.0057	0.61/0.0050	2.22/0.0105	1.18/0.0052	8.74/0.0425	3.03/0.0171	**15.76/0.1485**†
`ibm_amlsim_li_medium`	23.19/0.1811	22.40/0.1718	23.55/0.1799	18.24/0.1270	3.41/0.0143	2.30/0.0073	4.05/0.0211	2.30/0.0083	4.29/0.0238	18.74/0.1134	5.02/0.0266	**23.38/0.2077**†
`data_generator`	100.00/1.0000	100.00/1.0000	99.99/1.0000	100.00/1.0000	99.74/0.9979	99.63/0.9956	99.98/1.0000	100.00/1.0000	62.72/0.6327	100.00/1.0000	82.72/0.9923	**99.93/1.0000**†
`dgraphfin`	98.23/0.9978	98.45/0.9980	98.59/0.9989	97.60/0.9873	2.60/0.0129	3.26/0.0161	2.71/0.0123	3.25/0.0189	2.59/0.0094	96.55/0.9431	0.42/0.0114	**97.91/0.9979**†
`cc_transactions`	51.15/0.5126	52.26/0.5092	52.76/0.5184	49.51/0.4967	48.16/0.3472	49.24/0.3555	48.62/0.3441	48.74/0.3883	4.79/0.0215	52.15/0.5085	48.24/0.4167	**51.40/0.5213**†
Macro-Average	67.92/0.6799	67.14/0.6666	66.62/0.6917	64.05/0.6356	23.78/0.2143	24.31/0.2170	23.33/0.2011	25.72/0.2353	18.94/0.1480	55.46/0.5223	14.54/0.1691	**67.26/0.6848**†

TABLE S4

Comparative Evaluation of Conformal Risk Control against Calibration and Conformal Baselines on Elliptic-v1 (Target Significance $\alpha = 0.01$, Nominal Coverage $1 - \alpha = 99.0\%$).

Calibration / Conformal Method	Overall Cov. (%)	Benign Cov. (%)	Illicit Cov. (%)	Illicit Cov. Error	Mean Set Size $\mathbb{E}[\lvert\Gamma\rvert]$	Empty Sets (%)
Uncalibrated Cutoff ($\tau = 0.50$)	91.42	99.85	90.10	−8.90 pp	1.000	0.00%
Temperature Scaling [13]	94.20	99.70	92.80	−6.20 pp	1.000	0.00%
Isotonic Regression [14]	96.10	99.65	94.50	−4.50 pp	1.000	0.00%
Marginal Conformal Prediction (Split CP) [15]	99.05	99.98	82.40	−16.60 pp	1.001	0.00%
Class-Conditional CRC (Ours, Guarantee A)	**99.05**	**99.05**	**99.12**	**+0.12** pp	**1.005**	**0.00**%
Streaming Adaptive Conformal Inference (Ours, Guarantee C)	**99.04**	**99.06**	**99.08**	**+0.08** pp	**1.006**	**0.00**%

TABLE S5

Adversarial Camouflage Stress-Test on Elliptic-v1: Macro F1-Score (%) under Increasing Perturbation Ratio ρ.

Camouflage Ratio (ρ)	C-STGB (Ours)	CARE-GNN	Vanilla HGT	GCN
0.0 (Clean)	**91.42**	82.00	78.50	18.73
0.1	**91.39**	80.75	73.34	17.23
0.2	**91.31**	78.91	67.43	15.73
0.4	**90.97**	74.40	54.78	12.73
0.6	**90.41**	69.13	41.44	9.73
0.8	**89.63**	63.30	27.65	6.73

C. Disentangled Streaming Latency Profiling

Table S7 reports the itemized computational budget across individual pipeline stages on streaming transactions (CPU: AMD 5900X, GPU: RTX 3080). The lightweight Fast-Path achieves an end-to-end latency of 0.45 ms per transaction event, and batch-64 amortized throughput reaches 0.054 ms per transaction, fully satisfying production banking SLAs (< 35 ms).

Appendix F
Exhaustive Hyperparameter Tuning & Optimization Protocol

Model hyperparameter tuning for C-STGB is conducted via Tree-Structured Parzen Estimator (TPE) Bayesian optimization using Optuna across $N = 50$ trials per benchmark dataset on the training split $\mathcal{D}_{\text{train}}$. Table S8 reports the 18-parameter search space, prior distribution archetypes, and globally optimal calibrated parameters. The learning rate schedule incorporates AdamW with linear warm-up (5 epochs) followed by cosine annealing over $N_{\text{epochs}} = 100$ epochs with early stopping patience of 15 epochs based on validation Minority F1.

Appendix G
Graph Topological Metrics & Structural Sparsity

Table S6 documents fine-grained graph structural descriptors across all 14 benchmark networks. This structural profiling elucidates the performance dichotomy between relational graphs and tabular streams:

1) **High-Diameter Relational Graphs (Group A & B-Relational):** Datasets such as `elliptic_v1` ($D = 18, C = 0.048$), `ibm_amlsim_hi` ($D = 18, C = 0.068$), and `saml_d` ($D = 21, C = 0.034$) exhibit extended multi-hop directed diameters and significant clustering coefficients. In these environments, launderers execute smurfing and pass-through layering across multiple accounts, making spatio-temporal message aggregation strictly superior to isolated tabular models.
2) **Low-Diameter Bipartite Stars (Group B-Flat & Group C):** Conversely, `paysim1` ($D = 2, C = 0.000$) and `cc_transactions` ($D = 2, C = 0.000$) represent flat transaction streams where entities function as isolated star centers with zero multi-hop cycles. Graph convolutions over such bipartite trees induce uniform diffusion, explaining why tree ensembles excel on these specific benchmarks.

Appendix H
Fine-Grained Leave-One-Feature-Out Invariant Sensitivity Analysis

Table S9 isolates the marginal predictive utility of each individual invariant in $\mathbf{z}_{\text{inv}}(u, t) \in \mathbb{R}^{12}$ via leave-one-feature-out ablation across three diverse benchmarks. Flow Regularity ($\tilde{\Phi}_{\text{flow}}$) and Hawkes Burst Intensity (λ_u) represent the most critical single descriptors: dropping $\tilde{\Phi}_{\text{flow}}$ drops F1 by 4.52 pp

TABLE S6
FINE-GRAINED STRUCTURAL AND TOPOLOGICAL PROPERTIES OF THE 14 EVALUATED FINANCIAL TRANSACTION NETWORKS.

Dataset Identifier	Entities $\|\mathcal{V}\|$	Trans. $\|\mathcal{E}\|$	Directed Density ρ	Clustering C	Assortativity r	Diameter D	Horizon ΔT	Class Skew π
elliptic_v1	203,769	234,355	5.64×10^{-6}	0.048	−0.142	18	14 days	9.80%
elliptic_v2	122,283	427,330	2.86×10^{-5}	0.089	−0.118	14	14 days	6.18%
xblock_eth	2,978,658	3,485,913	3.93×10^{-7}	0.012	−0.185	26	365 days	0.04%
mtgox_leaked	119,343	148,822	1.04×10^{-5}	0.061	−0.094	12	120 days	0.82%
saml_d	288,574	345,188	4.15×10^{-6}	0.034	−0.162	21	180 days	0.38%
paysim1	6,362,620	6,362,620	1.57×10^{-7}	0.000	−0.002	2	30 days	0.13%
paysim_extended	1,048,575	1,048,575	9.53×10^{-7}	0.000	−0.001	2	30 days	0.12%
ibm_amlsim_hi_small	44,057	114,834	5.92×10^{-5}	0.072	−0.131	16	90 days	0.28%
ibm_amlsim_hi_medium	220,285	574,170	1.18×10^{-5}	0.068	−0.138	18	90 days	0.27%
ibm_amlsim_li_small	44,057	98,421	5.07×10^{-5}	0.045	−0.155	19	90 days	0.11%
ibm_amlsim_li_medium	220,285	492,105	1.01×10^{-5}	0.042	−0.159	22	90 days	0.11%
data_generator	50,000	120,000	4.80×10^{-5}	0.081	−0.120	15	60 days	0.50%
dgraphfin	1,210,092	4,085,410	2.79×10^{-6}	0.055	−0.174	24	180 days	0.61%
cc_transactions	284,807	284,807	3.51×10^{-6}	0.000	0.000	2	2 days	0.17%

TABLE S7
FINE-GRAINED LATENCY PROFILING PER TRANSACTION EVENT (CPU: AMD 5900X, GPU: RTX 3080).

Processing Stage	Mean Latency	P95 Latency	P99 Latency
1. DuckDB Ingestion & 12-D Invariant Extraction	0.11 ms	0.16 ms	0.22 ms
2. Dynamic Top-K ($K \leq 15$) Subgraph Retrieval	0.18 ms	0.26 ms	0.35 ms
3. Temporal GNN & Adaptive Fusion Forward Pass	0.12 ms	0.17 ms	0.21 ms
4. Conformal CRC Calibration & 3-Tier Routing	0.04 ms	0.05 ms	0.07 ms
Total End-to-End In-Flight Fast-Path	**0.45** ms	**0.64** ms	**0.82** ms
5. GNNExplainer Subgraph Extraction (Tier 1 Alerts)	1.65 ms	2.10 ms	2.85 ms
Total End-to-End Full-Path (with Attribution)	**2.10** ms	**2.74** ms	**3.67** ms
Batch-64 Amortized Throughput (GPU)	**0.054** ms	**0.072** ms	**0.095** ms

TABLE S8
HYPERPARAMETER SEARCH SPACE AND OPTIMAL CONFIGURATION GRID.

Module / Component	Hyperparameter	Search Interval / Set	Optimal Value
Optimization	Initial Learning Rate η	$[10^{-4}, 10^{-2}]$ (Log-uniform)	1.2×10^{-3}
Optimization	Weight Decay λ_{reg}	$[10^{-6}, 10^{-3}]$ (Log-uniform)	1.0×10^{-4}
Optimization	Batch Size B	$\{64, 128, 256, 512\}$	256
Spatio-Temporal	Hidden Dimension d	$\{64, 128, 256\}$	128
Architecture	GNN Layers L	$\{2, 3, 4\}$	2
Architecture	Dropout Rate p_{drop}	$[0.10, 0.50]$ (Uniform)	0.20
Architecture	Tri-Band Dim d_t	$\{16, 32, 64\}$	32
Architecture	Edge Subgraph Top-K	$\{5, 10, 15, 20\}$	15
Camouflage Defense	Edge Safety Floor δ_{floor}	$[0.05, 0.25]$ (Uniform)	0.10
Camouflage Defense	Auxiliary Loss Weight γ_2	$[0.01, 0.20]$ (Uniform)	0.10
Typology SMOTE	Latent Clusters K	$\{2, 3, 4, 6\}$	4
Typology SMOTE	Oversampling Ratio ρ_{syn}	$[0.20, 1.00]$ (Uniform)	0.50
Typology SMOTE	Bilinear Link Thresh τ_{edge}	$[0.40, 0.80]$ (Uniform)	0.65
Typology SMOTE	Reconstruction Weight γ_1	$[0.10, 1.00]$ (Uniform)	0.50
Loss Function	Tversky Weight α_{tv}	$[0.50, 0.90]$ (Uniform)	0.70
Loss Function	Tversky Weight β_{tv}	$[0.10, 0.50]$ (Uniform)	0.30
Loss Function	Focal Exponent γ_{focal}	$[1.0, 3.0]$ (Uniform)	2.0
Conformal Triage	Significance Level α	$\{0.005, 0.01, 0.02, 0.05\}$	0.01
Conformal Triage	ACI Adaptation Rate γ_{aci}	$[0.001, 0.05]$ (Log-uniform)	0.005
Conformal Triage	Cost Ratio $C_{\text{FN}}/C_{\text{FP}}$	$[5.0, 30.0]$ (Uniform)	15.0

on elliptic_v1 and 5.80 pp on ibm_amlsim_hi, confirming that money laundering fundamentally exploits mass-flow conservation through intermediate mules. Similarly, removing causal historical taint propagation ($\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$) incurs a 3.89 pp penalty on crypto ledgers.

TABLE S9
LEAVE-ONE-FEATURE-OUT SENSITIVITY ANALYSIS ON 12-D INVARIANTS $\mathbf{z}_{\text{INV}}$ (MINORITY F1-SCORE (%) / PR-AUC).

Invariant Configuration	Elliptic-v1	IBM-AMLSim HI	SAML-D
Full 12-D Descriptor Vector $\mathbf{z}_{\text{inv}}$	**99.44/1.0000**	**37.70/0.3550**	**93.68/0.9574**
w/o Flow Regularity $\tilde{\Phi}_{\text{flow}}$	94.92/0.9610	31.90/0.2980	88.12/0.9015
w/o Degree Asymmetry Ψ_{asym}	98.10/0.9880	35.40/0.3310	91.80/0.9380
w/o Hawkes Burst Intensity λ_u	96.35/0.9725	33.15/0.3095	89.50/0.9120
w/o Causal Fwd Taint $\mathbf{s}_{\text{fwd}}$	95.55/0.9680	34.80/0.3240	90.75/0.9250
w/o Causal Bwd Taint $\mathbf{s}_{\text{bwd}}$	96.80/0.9790	35.10/0.3280	91.40/0.9310
w/o Amount Mean μ_A	98.80/0.9940	36.20/0.3410	92.60/0.9460
w/o Amount Variance σ_A^2	98.50/0.9915	36.05/0.3395	92.35/0.9430
w/o Amount Skewness γ_A	99.10/0.9970	36.90/0.3480	93.10/0.9510
w/o Amount Kurtosis κ_A	99.15/0.9975	37.05/0.3490	93.20/0.9520
w/o Amount Velocity ν_A	97.90/0.9850	34.60/0.3220	91.10/0.9290
w/o IO Velocity Ratio ν_{io}	98.30/0.9890	35.80/0.3360	92.05/0.9405
w/o Inter-Arrival Dispersion $\sigma_{\Delta t}$	97.40/0.9810	34.10/0.3180	90.60/0.9230

APPENDIX I
REGULATORY COMPLIANCE, AUDITABLE TRIAGE SOP & GOVERNANCE PROTOCOL

Table S10 outlines the Standard Operating Procedure (SOP) governing the integration of C-STGB within production banking operations under the Bank Secrecy Act (BSA, 31 U.S.C. 5318(g)), Federal Reserve SR 26-2, and FATF Recommendation 16.

1) **Tier 1 (Automated Dossier Generation):** Confirmed high-risk transactions ($\Gamma(u) = \{1\}$) immediately trigger a temporary asset hold and generate a cryptographically signed FinCEN Form 111 XML filing package with automated multi-hop subgraph attribution. Filing occurs strictly within the statutory 30-day window following human BSA officer sign-off.
2) **Tier 2 (Human-in-the-Loop Queue):** Ambiguous alerts ($\Gamma(u) = \{0, 1\}$) are routed to specialized investigative queues. If supplementary KYC/customer documentation confirms benign origin within 72 hours, the transaction is released without filing.
3) **Tier 3 (Straight-Through Clearing):** Transactions classified as decisive negatives ($\Gamma(u) = \{0\}$) bypass human intervention with a provable finite-sample error rate $\leq 0.12\%$.

A. Enterprise Streaming Architecture & SLA Guarantees

In enterprise deployments, C-STGB integrates into event streaming message brokers (Apache Kafka / Flink) via zero-copy Apache Arrow buffers. Incoming transaction events populate rolling DuckDB state tables ($t_e \leq t$), updating 1-hop and 2-hop ego-neighborhoods within 0.11 ms. The multi-threaded PyTorch C++ inference engine evaluates forward message passing and Class-Conditional Conformal bounds in

0.34 ms, guaranteeing that 99.9% of transactions complete processing within $<$ 1.0 ms, substantially exceeding Tier-1 banking latency SLAs ($<$ 35 ms).

B. Model Risk Management (MRM) Protocol under SR 11-7 / SR 26-2

In compliance with supervisory guidance on Model Risk Management (Federal Reserve SR 11-7 and SR 26-2), institutions deploying C-STGB execute continuous rolling monitoring:

1) **Quarterly Backtesting:** Non-conformity calibration sets $\mathcal{D}_{\text{cal}}$ are updated on a 90-day rolling window to recalibrate non-conformity quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$.
2) **Drift Detection & Circuit Breakers:** If the empirical Total Variation drift metric $d_{\text{TV}}(\mathcal{P}_t, \mathcal{P}_{\text{cal}})$ exceeds $\epsilon_{\text{drift}} > 0.05$, the system triggers an automated alert, temporarily expanding Tier 2 review bandwidth until recalibration converges.
3) **Challenger Benchmark Shadowing:** High-capacity tree ensembles (XGBoost/CatBoost) execute in parallel as shadow challenger models to detect unexpected divergence on isolated single-hop transactions.

TABLE S10
REGULATORY ESCALATION MAPPING ACROSS CONFORMAL TRIAGE TIERS.

Operational Dimension	Tier 1 (Escalation)	Tier 2 (Queue)	Tier 3 (Clearance)
Prediction Set $\Gamma(u)$	$\{1\}$ (Singleton)	$\{0,1\}$ (Abstain)	$\{0\}$ (Singleton)
Risk Category	Confirmed Illicit	Boundary Ambiguity	Confirmed Benign
Action Executed	Quarantined Hold	Compliance Queue	Straight-Through
Statutory Timeframe	Immediate Hold	72-Hour Review	Real-Time ($<$ 1 ms)
Regulatory Output	FinCEN Form 111	Examiner Audit Log	Clearing Receipt
Human Involvement	Mandatory Sign-Off	Full Investigation	Zero (Automated)
Error Budget	False Discovery $<$ 1%	N/A (Selective)	False Neg. $\leq$ 0.12%

APPENDIX J
REPRODUCIBILITY & OPEN SCIENCE SPECIFICATION

Table S11 provides the complete software, hardware, and environment specification. All secondary experimental artifacts—including the 14-dataset hardware memory profiling, the 18-parameter hyperparameter configuration grid, the leave-one-out 12-D invariant ablation, the 8-pair zero-shot transfer matrix, and the multi-agent AST grammar schema—are openly available in the project repository: https://github.com/NazmulHasanNihal/Intelligent-AML.

TABLE S11
REPRODUCIBILITY ENVIRONMENT AND SOFTWARE STACK.

Reproducibility Dimension	Specification / Value
Code Repository	https://github.com/NazmulHasanNihal/Intelligent-AML
Primary Programming Language	Python 3.10.12
Deep Learning Framework	PyTorch 2.3.0+cu121, PyTorch Geometric (PyG) 2.5.2
Graph Kernel Engine	DGL 2.2.1, NetworkX 3.2.1
Relational Data Engine	DuckDB 0.10.2, Apache Arrow 16.0.0
Scientific Stack	NumPy 1.26.4, SciPy 1.13.0, Scikit-learn 1.4.2
Conformal Prediction Library	MAPIE 0.8.0, Custom Class-Conditional CRC Engine
Hardware Environment	Kaggle Cloud TPU v3-8 / NVIDIA Tesla T4 GPU (16GB)
Workstation Environment	AMD Ryzen 9 5900X, 64GB DDR4 RAM, RTX 3080 (10GB)
Operating System	Ubuntu 22.04 LTS / Windows 11 Enterprise (x86_64)
Random Seeds Evaluated	$seeds \in \{42, 101, 2024, 7, 999\}$
Master Training Command	`python scripts/train_unified_pipeline.py --dataset all`
Benchmark Evaluation Script	`python scripts/benchmark_baselines.py --runs 5`
Figure Generator Script	`python scripts/generate_all_publication_figures.py`

REFERENCES

[1] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
[2] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018.
[3] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 3146–3154.
[4] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *International Conference on Learning Representations (ICLR)*, 2017.
[5] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *International Conference on Learning Representations (ICLR)*, 2018.
[6] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in *International Conference on Learning Representations (ICLR)*, 2019.
[7] A. Pareja, G. Domeniconi, J. Chen, T. Teng, T. Wang, K. Chai, M. Magdon-Ismail, and C. Shen, "EvolveGCN: Evolving graph convolutional networks for dynamic graphs," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 04, 2020, pp. 5363–5370.
[8] E. Rossi, B. Zhou, F. Monti, F. Frasca, U. Alon, and M. M. Bronstein, "Temporal graph networks for deep learning on dynamic graphs," in *NeurIPS Workshop on Graph Representation Learning and Beyond*, 2020.
[9] D. Xu, C. Ruan, E. Korpeoglu, S. Kumar, and K. Achan, "Inductive representation learning on temporal graphs," in *International Conference on Learning Representations (ICLR)*, 2020.
[10] Y. Dou, Z. Liu, L. Sun, Y. Deng, H. Peng, and P. S. Yu, "Enhancing graph neural network-based fraud detectors against camouflaged fraudsters," in *Proceedings of the 29th ACM International Conference on Information & Knowledge Management (CIKM)*, 2020, pp. 315–324.
[11] T. Zhao, X. Zhang, and S. Wang, "GraphSMOTE: Imbalanced node classification on graphs with graph neural networks," in *Proceedings of the 14th ACM International Conference on Web Search and Data Mining (WSDM)*, 2021, pp. 833–841.
[12] I. Gibbs and E. J. Candès, "Adaptive conformal inference under distribution shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 1660–1672.
[13] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330.
[14] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002, pp. 694–699.
[15] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer Science & Business Media, 2005.